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Physics of Fluids 33, 095116 (2021)

<https://doi.org/10.1063/5.0062377>



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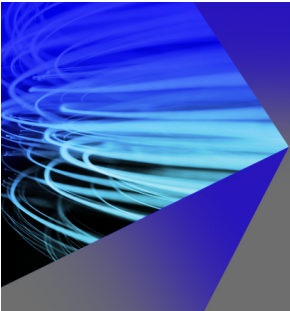
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Cite as: Phys. Fluids **33**, 095116 (2021); doi: [10.1063/5.0062377](https://doi.org/10.1063/5.0062377)

Submitted: 5 July 2021 · Accepted: 27 August 2021 ·

Published Online: 15 September 2021



View Online



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ABSTRACT

Gaining and understanding flow dynamics have much importance in a wide range of disciplines, e.g., astrophysics, geophysics, biology, mechanical engineering, and biomedical engineering. For turbulent flows, local flow information such as velocity and its statistics, can be measured experimentally. Due to the poor fidelity or experimental limitations, some information may not be resolved in a region of interest. On the other hand, detailed flow features are described by the governing equations, e.g., the Navier–Stokes equations for viscous fluid, and can be resolved numerically, which is heavily dependent on the capability of either computing resources or modeling. Alternatively, we address this problem by employing the physics-informed deep learning and treat the governing equations as a parameterized constraint to recover the missing flow dynamics. We demonstrate that with limited data, no matter from experiment or others, the flow dynamics in the region where the required data are missing or not measured, can be reconstructed with the parameterized governing equations. Meanwhile, a richer dataset, with spatial distribution of the control parameter (e.g., eddy viscosity of turbulence modelings), can be obtained. The method provided in this paper may shed light on the data-driven scale-adaptive turbulent structure recovering and understanding of complex fluid physics and can be extended to other parameterized governing systems beyond fluid mechanics.

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I. INTRODUCTION

In fluid mechanics, acquiring the flow data is one of the key tasks in science and engineering.^{1,2} Analyzing or controlling the flow dynamics generally needs a rich dataset or a continuous representation of the flow field. Conventionally, the flow dynamics is often conducted by either numerics or experiments. Numerically, the Navier–Stokes equations (NSEs), which govern the flow dynamics in a form of partial differential equations (PDEs), can be solved under typical assumptions or specific conditions within a typical range of precision. For high Reynolds number flows, or turbulent flows, directly solving the NSEs is extremely challenging due to limited computer resources and large amounts of degrees of freedom.^{2,7,13} To eliminate the restriction, various turbulence models, which are coupled with the NSEs, are proposed to save computational cost.³ Meanwhile, suspicion around correctness and

applicability of modeling arise, which lead to unremitting pursuit of research in turbulence modeling,^{8,25,28} although machine learning approaches have been used to enhance the predictive capability.²⁴

In contrast, nowadays, gain of reliable data from experiments generally has no restriction from the Reynolds number. However, due to the precision and uncertainty of experimental equipment,²⁸ measured data can be obtained in a finite set of locations and are precise only in a limited-order statistical sense, which are not as sufficiently accurate as the solutions described by the NSEs. In a statistical sense, the experimental data can be generally regarded as an approximation of the solutions of the NSEs. It turns out that leveraging the limited experimental data to the underlying governing equations appears as a critical question in modeling and recovering the flow dynamics. In order to address this problem, we here consider the reformed governing equations where the control parameter is regarded as an unknown

and employ the physics-informed learning technique to preserve fundamental physical principles,^{20,22,24} by which the parameterized governing equations and the missing flow dynamics can be physically determined at the same time. To achieve the above, the parameterized governing equations are treated as a physical constraint, which is used to form a loss function in the training of neural networks.¹⁶

A data-driven discovery of PDEs, named PDE-FIND, was proposed by Rudy and his collaborators. Especially, for the NSEs, the PDE-FIND was used to identify the Reynolds number in the vorticity transport equation for fluid mechanics.²¹ The research group of Karniadakis has made a great contribution in physics-informed neural networks^{11,12,16} and their applications to fluid flows. Generally, physics-informed neural networks can be used for both forward and inverse problems. For forward problem, Raissi *et al.* proposed the hidden fluid mechanics,²⁰ which provide a new way to learn velocity and pressure fields from flow visualizations. Most recently, Jin *et al.* proposed the NSFNETs (Navier-Stokes-Flow nets) to solve two different, velocity-pressure and vorticity-velocity, formulations of the incompressible NSEs, in the sense of direct numerical simulation for Kovasznay flow and turbulent channel flow.⁶ Other applications are for high-speed inviscid flow⁹ and vortex-induced vibration.¹⁹ For inverse problems, physics-informed neural networks were used to identify the Reynolds number from flow data for the low Reynolds number flow past a cylinder¹⁶ and to discover universal variable-order fractional model for turbulent Couette flow.¹⁰

Built on the above evidence, the generalization of physics-informed neural networks forms the basis of the parameterized governing system in exploring missing flow physics and exploiting the essential of modeling parameters, in particular, for turbulent flows. To overcome numerical stiffness, a recently developed learning rate annealing algorithm²⁶ is adopted in this work. The proposed idea can be extended to a broad range of applications where PDEs appear as physical constraints, and the control parameter of the PDEs plays a role in dominating the physics.

II. METHODOLOGY

Mathematically, we consider the following system as a general description of physical phenomena in a region Ω :

$$\mathcal{L}^t(\vec{p}; \vec{p}_0, \vec{g}_\Gamma, \nu) = \vec{f}, \quad (1)$$

where the operator $\mathcal{L}^t$ represents the evolution of $\vec{p}$ and $\vec{p}_0$ is the initial condition. $\vec{g}_\Gamma$ is the boundary conditions where Γ denotes the boundary $\partial\Omega$ of Ω . ν denotes the control parameter. For a steady system, the corresponding representation can be written as $\mathcal{L}^\circ(\vec{p}; \vec{g}_\Gamma, \nu) = \vec{f}$. In real-world applications, in the sense of temporal (or ensemble) *averaging* or spatial *filtering*, the control parameter can be represented as an effective one, ν_{eff} , which is the summation of the physical ν and the modeled ν_b , while the original expression of the governing system is kept the same. For instance, in turbulence modeling, at any level of turbulence, closure of the averaged or filtered NSEs requires the parameter, eddy(subgrid) viscosity, which is introduced in large eddy simulations (LES), scale-resolving simulations (SRS), and Reynolds-averaged Navier-Stokes (RANS) methods.^{3,23} Once the effective control parameter ν_{eff} is introduced as an unknown parameter, $\mathcal{L}^t(\vec{p}; \vec{p}_0, \vec{g}_\Gamma, \nu_{\text{eff}}) = \vec{f}$ or $\mathcal{L}^\circ(\vec{p}; \vec{g}_\Gamma, \nu_{\text{eff}}) = \vec{f}$ can be regarded as the parameterized physical constraint, which is used to explore missing flow dynamics under the framework of the physics-informed neural networks.

Let us consider the statue of Leonardo shown in Fig. 1(a), as a physical field. It is assumed that a part of Leonardo's mustache is missing. In order to fix the statue, we need to find a pattern that is consistent with the missing one. Without referring the potential governing system of the statue, by learning the mustache pattern beyond the information-missing region, it is possible to create a pattern that matches with the pattern around the region. It is clear that in principle, there should exist infinite possibilities. So, *by virtue of limited information, how to physically make it unique as possible as we can?* This is termed as “the parameterized physics-informed deep learning strategy.”

As illustrated in Fig. 1(b), $\vec{p}$ is known on $\Omega \setminus \Omega'$ and unknown on Ω' . Theoretically, in fluid mechanics, if the effective eddy viscosity ν_{eff} can be estimated or physically formulated, the missing $\vec{p}$ on Ω' can be solved from Eq. (1). In the framework of the parameterized physics-informed learning, solving Eq. (1) is avoided, and Eq. (1) only leads us to directly find a physical unique solution on Ω' in terms of $\vec{p}$ on $\Omega \setminus \Omega'$. Let us turn back to the physical world of the fluid dynamics, which is governed by the PDEs, e.g., the incompressible NSEs as follows:

$$\mathcal{L}^t(\{\mathbf{u}, p\}; \mathbf{g}, \nu) \equiv \begin{cases} \frac{\partial \mathbf{u}}{\partial t} - \nu \nabla^2 \mathbf{u} + \mathbf{u} \cdot \nabla \mathbf{u} + \nabla p = \mathbf{f}, & \text{on } \Omega, \\ \nabla \cdot \mathbf{u} = 0, \\ \mathbf{u}|_{\Gamma=\partial\Omega} = \mathbf{g}. \end{cases} \quad (2)$$

Theoretically, Eq. (2) can be used for all kinds of incompressible flows. However, considering the fixed reference length scale and velocity magnitude, with extremely small viscosity ν and, thus, very high Reynolds number, solving Eq. (2) is extra challenging in practical applications. In another aspect, although some data can be obtained experimentally, it is also challenging, if possible, to establish the direct connection between Eq. (2) and the gained experimental data. In engineering applications, the RANS equations, which are derived by the Reynolds decomposition of flow fields, have been well validated for engineering applications by experimental data and direct simulations of the NSEs.^{3,23} The key question in RANS is to model the Reynolds stress τ_{ij} , which is represented as follows:

$$\tau_{ij} = -\rho \overline{u'_i u'_j}, \quad (3)$$

where $\overline{(\cdot)}$ denotes the Reynolds average or the spatial filtering and $u'_i = u_i - \bar{u}_i$. Similar to Fick's law, $\rho \overline{u'_i u'_j}$ can be modeled as

$$\rho \overline{u'_i u'_j} = -\rho \nu_t \left(\frac{\partial \bar{u}_i}{\partial x_j} + \frac{\partial \bar{u}_j}{\partial x_i} \right), \quad (4)$$

where ν_t denotes the eddy viscosity. Although it is named as viscosity, ν_t is a parameter different from the molecular viscosity in the original equations. Similar idea has been adopted to the large eddy simulation (LES), where sub-grid eddy viscosity ν_{sgs} is introduced to model the subgrid scale turbulent flow. For hybrid RANS/LES, such as detach eddy simulation (DES) and scale adaptive simulation (SAS), ν_t in the control system has different meaning in different region of the computational domain, depending on the resolution of the turbulence scales. Normally, ν_t is flow dependent. In the turbulence community, various modeling of the parameter ν_t has been studied for several decades. With the big data and machine learning, model constants of the RANS equations are tuned by fitting.^{2,4,8,15} However, compared with

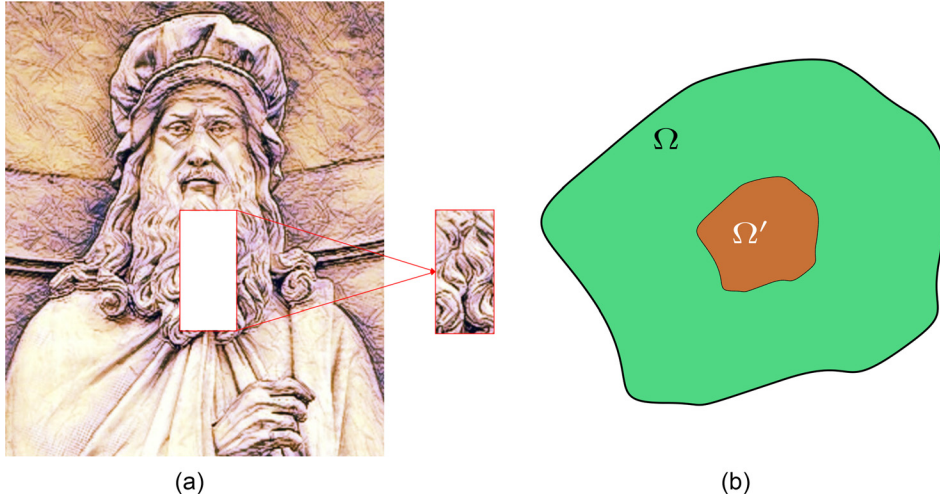


FIG. 1. Illustration of missing information: (a) statue of Leonardo (photograph by the first author H.X.) with a part of curly mustache missing; (b) mathematical representation of the domain Ω and the sub-domain Ω' on which the information is missing.

small datasets, observed discrepancies are not explained any better with big ones⁴ and modeling ν_t , so as the ν_{sgs} in a universal sense still remains as a persistent challenging because of the case-by-case dependence. Luckily, engineering practices demonstrate that predictions of the RANS equations for complex flows are mostly quite satisfactory. It indicates that utilization of ν_t as an extra parameter can be a reliable way to establish the fitting between averaged/filtered experiment data and solutions of the corresponding control equations.

As indicated in the pioneering work,³ representation of turbulent transport by an eddy viscosity underlies most of the related theoretical work. With this idea, the classical RANS equations can be rewritten as (without considering the auxiliary equations)

$$\mathcal{L}'(\{\bar{\mathbf{u}}, p\}; g, \nu + \nu_\eta) \equiv \begin{cases} \frac{\partial \bar{\mathbf{u}}}{\partial t} - (\nu + \nu_\eta) \nabla^2 \bar{\mathbf{u}} + \bar{\mathbf{u}} \cdot \nabla \bar{\mathbf{u}} + \nabla \bar{p} = \mathbf{f}, & \text{on } \Omega, \\ \nabla \cdot \bar{\mathbf{u}} = 0, \\ \bar{\mathbf{u}}|_{\Gamma=\partial\Omega} = \mathbf{g}, \end{cases} \quad (5)$$

where ν_η is an artificial viscosity term introduced here for parameterization similar to that described above. From the control parameter point of view, the data-based determination of ν_η indicates “scale adaption” of recovering turbulent structures, same as ν_t in DES or SAS. The above equation is the system of the operator equation, Eq. (1), in a concrete way. For steady cases, the unsteady term $\partial \bar{\mathbf{u}} / \partial t$ is neglected. It should be mentioned that reformulating the NSEs does not imply employment of the RANS approach since the parameterization of $\nu_{\text{eff}} = \nu + \nu_\eta$ appears only as an undetermined parameter. The parameterization is significantly distinguished from the conventional tuning of model constants. The question remains as “how to determine the parameter ν_{eff} (or ν_η) regarded as a spatial distributed variable?”

With the parameterized NSEs, assuming only limited data are available on a set $\mathcal{X} \in \Omega \setminus \Omega'$ (cardinality $\mathcal{N} = |\mathcal{X}|$) of finite scattered measurement locations, we are interested in the missing flow dynamics on Ω' , where the specified point set $\mathcal{X}'$ with cardinality $\mathcal{N}' = |\mathcal{X}'|$ is chosen to represent the dynamics. On $\mathcal{X}$, the set of measured

averaged/filtered $\mathbf{u}$ is denoted by $\bar{\mathcal{S}}_{\mathcal{X}}$, and on Ω' , the set of missing averaged/filtered $\mathbf{u}$ is denoted by $\bar{\mathcal{S}}_{\mathcal{X}'}$. Furthermore, the undetermined ν_{eff} appears as a spatial variable defined on Ω and will be inferred later. Here, the target is to approximate the functional relation $(t, x, y, z) \mapsto (u, v, w, p, \nu_{\text{eff}})$ by physics-informed neural network $(t, x, y, z) \mapsto (e_u, e_v, e_w, e_{\text{div}})$. The parameterized NSEs are encoded in e_u, e_v, e_w and e_{div} , which represent the residuals of the momentum equations and the continuity equation, respectively. To overcome the numerical stiffness that leads to unbalanced back-propagated gradients during model training, a recently developed learning rate annealing algorithm is adopted.²⁶ The physics-informed residuals e_u, e_v, e_w and e_{div} are minimized with the mean squared error loss function on the time set $\mathcal{T}$,

$$\text{Loss} = \mathbb{E}_{\mathcal{T}} [\text{MSE}_{\text{Data}} + \text{MSE}_{\text{PDEs}}], \quad (6)$$

where

$$\text{MSE}_{\text{Data}} = \mathbb{E}_{\mathcal{X}} [||\mathbf{u}(t; \mathcal{X}) - \mathbf{u}_{(t; \mathcal{X})}||^2], \quad (7)$$

and

$$\text{MSE}_{\text{PDEs}} = \mathbb{E}_{\hat{\mathcal{X}}} [e_u^2 + e_v^2 + e_w^2 + e_{\text{div}}^2]. \quad (8)$$

Here, $\hat{\mathcal{X}} \in \Omega$. MSE_{Data} and MSE_{PDEs} denote “Mean Squared Errors” from data differences on the training dataset $\mathcal{X}$ and residuals of governing equations on the dataset $\hat{\mathcal{X}}$, respectively. The operator $\mathbb{E}_{\#}[\cdot]$ means the average on the set “#” in a distribution sense. $\mathbb{E}_{\mathcal{T}}[\cdot]$ is an identity operator for stationary modelings. Generally, the location cluster can be generated in a probability sense. Once the above minimization problem is solved, the extra-variables p and ν_{eff} can be discovered.

III. NUMERICAL CONFIGURATION

The diagram of the physics-informed neural network is illustrated in Fig. 2. We here treat the parameter ν_η as one output of the deep neural network. During the training process, the required weights of neural network are determined by the loss function formed

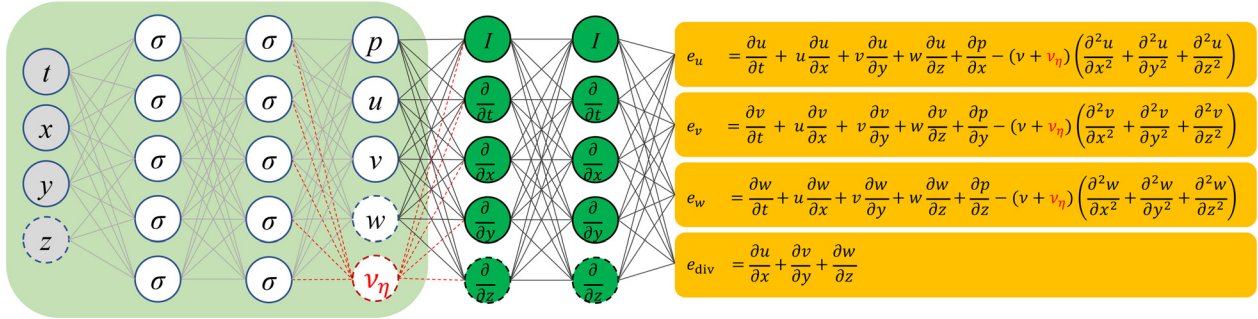


FIG. 2. Illustration of the physics-informed neural networks. The field variables $\{u, v, w, p\}$ and the parameter ν_η are regarded as an output variable of the neural network. σ represents the sinusoidal activation function used in this work.

by the residual of the governing equations. Once the training convergence is achieved, the variables $\{u, v, w, p, \nu_\eta\}$ can be inferred.

The training dataset was generated by computational fluid dynamics (CFD) with the open-source software OpenFOAM.²⁷ On the computational domain, unstructured hexhedras were used to partition the domain. For the governing equations, the convection terms and the time-derivative terms were discretized by the second-order upwind scheme and the second-order backward scheme, respectively. The PIMPLE algorithm, which is a combination of the PISO (Pressure Implicit with Splitting of Operator) and the SIMPLE (Semi-Implicit Method for Pressure-Linked Equations), is used to resolve the governing equations. In two dimensions, only values of the velocity components u and v at mesh nodes are considered as training data.

To check the ability of the method, we considered three different cases, related to the parameterized governing equations of RANS, unsteady RANS (URANS), and SAS for two different turbulent flows. SAS denotes “scale-adaptive simulation”²³ and typically allows simulation of unsteady flows with both RANS and LES contents. It should be noted that only regional velocity is used as training data, while velocity, pressure, and artificial viscosity in the entire domain are inferred. Furthermore, in order to validate availability of the proposed method for experimental data, artificial white noise (WN) is considered, which is added to the data generated by the mentioned CFD methods with relative amplitudes 1% and 10%, as extensive investigations.

Two kinds of flows, including a steady turbulent flow over a backward facing step (RANS) in Sec. IV A and an unsteady turbulent flow past a cylinder (URANS and SAS) in Secs. IV B and IV C, were involved. For the steady one, there are 1.2×10^4 cell nodes in total, while the random sampling nodes are 3×10^3 for training data (on $\Omega \setminus \Omega'$) and 4.3×10^3 constraints of the parameterized governing system (on Ω). A neural network with six hidden layers and 50 neurons per layer is considered. Both Adam and L-BFGS-B (limited-memory and bound constrained version of Broyden-Fletcher-Goldfarb-Shanno) optimizers are used simultaneously for the training.^{17,18} 2×10^5 epochs are used. For the unsteady flow past a cylinder, there are 351 snapshots and 5.1×10^5 cell nodes on every snapshot, thus 1.7901×10^8 in total. The random sampling nodes are 1.2×10^7 and 1.4×10^7 on all time snapshots. Corresponding neural network has 10 hidden layers and 96 neurons per layer. Adam is adopted for training. The learning rate is gradually decreased from 1.0×10^{-3} to 1.0×10^{-6} with 400 epochs at each stage.¹⁹ In Sec. IV C, the dataset with added white noise is used to mimic experimental condition during the training process.

IV. RESULTS

A. Steady case

We first consider a steady turbulent flow over a backward facing step, experimentally studied by Pitz and Daily date back to 1981.¹⁴ The Reynolds number is 5×10^5 based on the step height. The SST $k - \omega$ model was adopted to generate the data. Here, SST denotes “shear-stress transport.”²³ The contours of velocity components along the streamwise and vertical directions are given in Figs. 3(a) and 4(a), respectively. It is assumed that in the dashed-lined rectangular box, denoted by Ω' , the flow information therein is missing and, thus, not used as training data.

After the neural network is trained with the data outside Ω' , together with constraints of the governing equations, the flow variables in the whole domain Ω are inferred. Considering the reference CFD result, the L^2 errors of the inferred u and v are around 6% and 10%, respectively. In Figs. 3(b) and 4(b), the inferred velocity components by the physics-informed deep learning are given. It is clear that in the box region, we get the similar velocity fields as illustrated in Figs. 3(a) and 4(a), demonstrating that the averaged flow dynamics is mimicked correctly with the proposed parameterized NSEs. In Fig. 5, the inferred pressure field is in good agreement with the reference one from the SST $k - \omega$ model. It is interesting, on the other hand, to compare the inferred artificial viscosity ν_η with eddy viscosity ν_t empirically determined by the SST $k - \omega$ model. Although they have different meaning, thus are different in both magnitude and spatial distribution, as shown in Fig. 6, their distributions are related to the flow structure, and both are relatively large in the region with separation bubble.

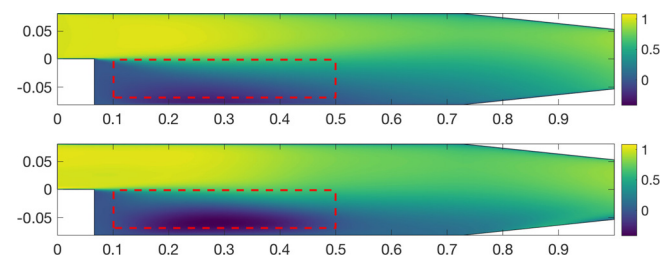


FIG. 3. Contours of the velocity component u in the streamwise direction for the steady turbulent flow over a backward facing step. (Top) From the SST $k - \omega$ model; (Bottom) inferred by the physics-informed neural network. The dashed-lined box indicates the region where the flow information is missing.

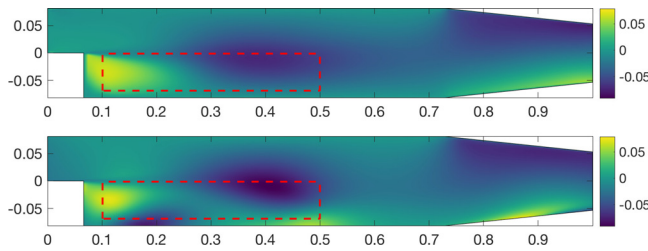


FIG. 4. Contours of the velocity component v in the vertical direction for the steady turbulent flow over a backward facing step. (Top) From the SST $k - \omega$ model; (Bottom) inferred by the physics-informed neural network. The dashed-line box indicates the region where the flow information is missing.

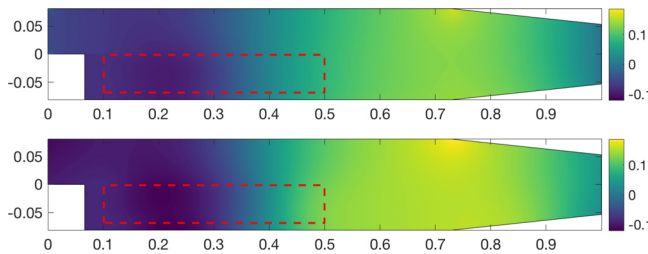


FIG. 5. The pressure fields for the steady turbulent flow over a backward facing step. (Top) From the SST $k - \omega$ model; (Bottom) inferred by the physics-informed neural network.

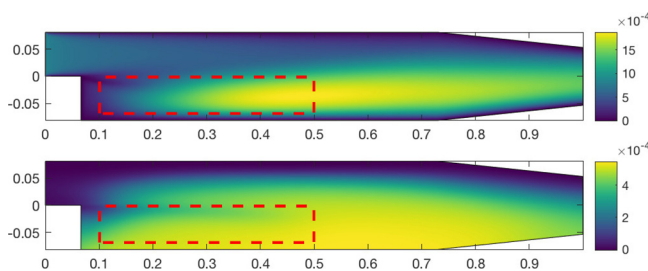


FIG. 6. Comparison between the eddy viscosity and the artificial viscosity for the steady turbulent flow over a backward facing step. (Top) Eddy viscosity ν_t from the SST $k - \omega$ model; (Bottom) artificial viscosity ν_η inferred by the physics-informed neural network.

For the same steady turbulent flow, we, in particular, consider a case where the parameter ν_η is removed, and thus, the original NSEs are used as a physics constraint during the learning. The energy contours of the inferred velocity field $\mathbf{u}_{\text{NSEs}}$ and the “perturbation” field ($\mathbf{u}_{\text{NSEs}} - \bar{\mathbf{u}}$) are given in Fig. 7. Special structures are observed around the tip of the separation bubble. Compared with the solution based on the parameterized NSEs, this significant inconsistency provides some potential information about the region where the most unstable modes or the energetic turbulent structures emerge around the flow separation region.

B. Unsteady case

As a further demonstration, we consider the time-dependent turbulent flow past a cylinder. The Reynolds number is

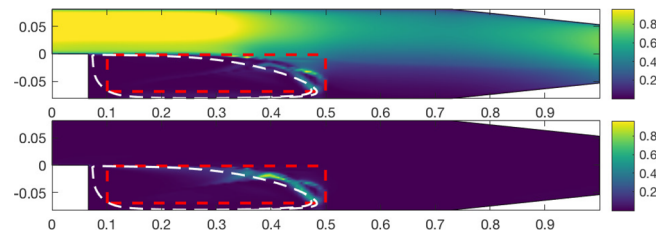


FIG. 7. The total energy and “perturbation” energy of the inferred velocity for the steady turbulent flow over a backward facing step. (Top) The inferred energy with the original NSEs; (Bottom) the “perturbation” energy. The white dashed-line indicates the separation region of the averaged field, while the red dashed-line box is the region where the flow information is missing.

$Re = u_{in}D/\nu = 10^5$, where u_{in} and D are the inlet velocity and diameter of the cylinder, respectively. The reference flows are generated by the SST $k - \omega$ and SST-SAS turbulence models. It should be emphasized that unlike other traditional RANS models (e.g., the SST $k - \omega$ model), the SST-SAS model has a pure RANS nature but achieves LES behavior. With this time-dependent turbulent flow, we demonstrate that the proposed methodology has a great adaptability to turbulence models for unsteady flow dynamics. In practice, the reference flow data obtained by the SST-SAS and SST $k - \omega$ turbulence models can be regarded as the “filtered” experimental data.

It is assumed, again, that in the system, there exists a local region where the flow dynamics is missing, as indicated by a dashed-rectangular box in Fig. 8. In order to employ the established methodology, we consider a series of flow field snapshots obtained by the SST-SAS model, and only a part of the velocity field beyond the dashed-rectangular box on every snapshot is employed for training. In Fig. 8(Mid), the reconstruction of the missing flow dynamics at $t = 11.96$ in the dashed-rectangular box is presented. Surprisingly, compared with Fig. 8(Top), the reconstructed missing flow coincides with the reference flow very well, even with complex vortex structure in the box. Qualitatively, we show the differences between the reference fields and the inferred ones in Fig. 8(Bottom). Relative low deviation is observed. Meanwhile, the profiles of the lift and drag coefficients are given in Fig. 9. The inferred C_L profile is consistent with the reference one. However, there exist some differences between the inferred C_D and the reference one. As observed from the comparison, the unsteadiness (or “irregularity”) of the C_D profile from the inferred data is reconstructed fairly well. Furthermore, we show the distribution of the viscosity in Fig. 10. It is observed that the patterns of the viscosity from both the SST-SAS and neural network are consistent with the vortex shedding pattern, but with different magnitudes. However, the inferred ν_η pattern is, in fact, different from the ν_t pattern in the turbulence modeling. This difference is possibly attributed to the definitions of ν_η and ν_t and their contributions to modeling flow structures.

Accordingly, for the flow past a cylinder, instantaneous results at $t = 7$ with the SST $k - \omega$ model are given in Figs. 11–13. We see that using the methodology with the physics-informed deep learning, the missing flow dynamics whose location is assigned arbitrary is recovered properly with good accuracy.

With the cases investigated, we demonstrate that when the parameter ν_η is introduced as an output variable of the neural network, the parameterized governing system of flow dynamics has a strong adaptivity to approximate the data from the turbulence models.

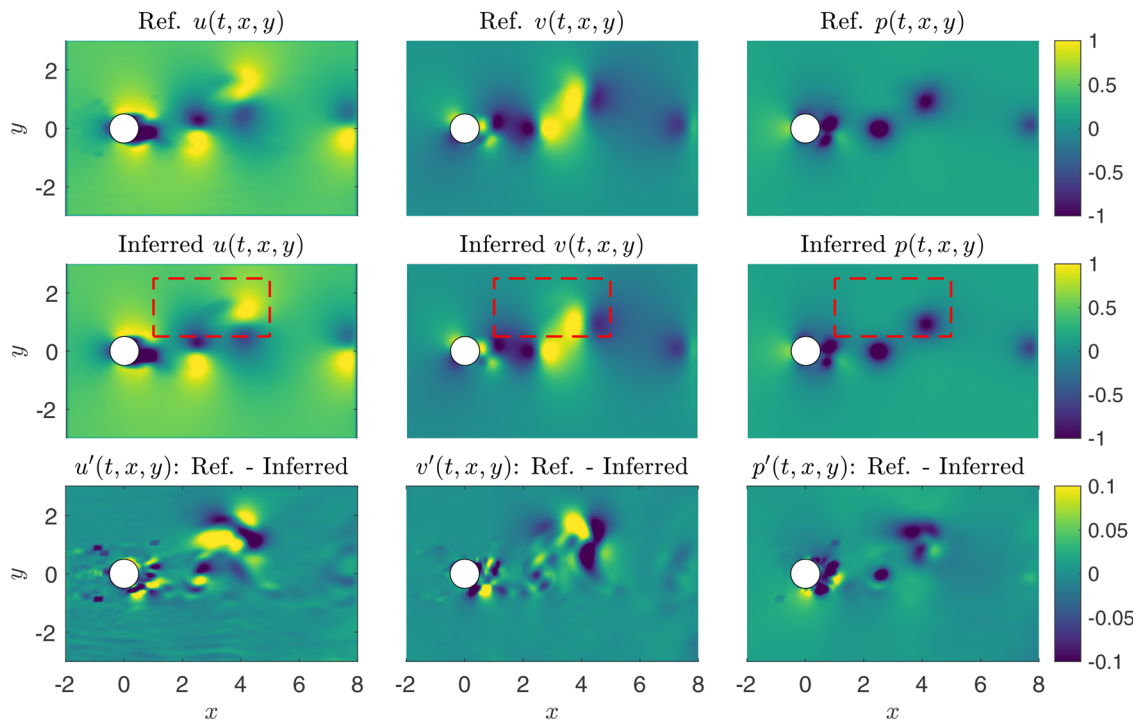


FIG. 8. Unsteady turbulent flow past a cylinder: (Top) The instantaneous reference velocity and pressure fields at $t = 11.96$, generated by the SST-SAS turbulence model; (Mid) the inferred fields at the same time instant with the missing dynamics; (Bottom) the difference between the reference fields and the inferred fields. The red dashed-line indicates the region where the flow dynamics is missing.

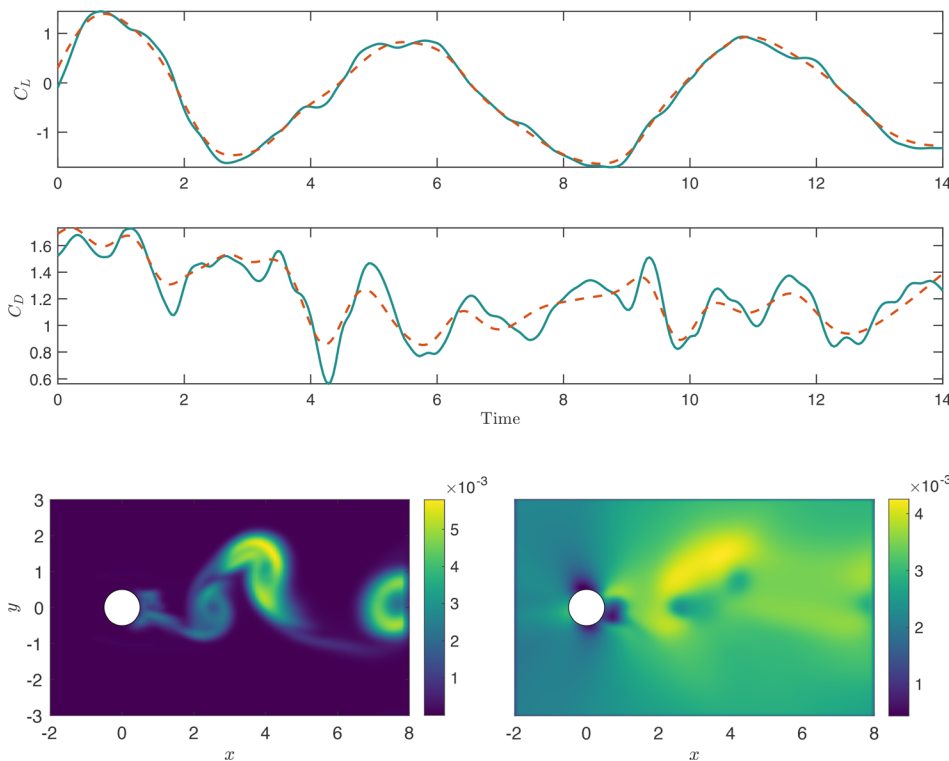


FIG. 9. Lift and drag coefficients over the cylinder: (Top) profiles of the lift coefficient; (Bottom) profiles of the drag coefficient. The solid curves are generated from the reference data by the SST-SAS turbulence model, while the dashed curves generated from the inferred fields.

FIG. 10. Comparison between the eddy viscosity and the artificial viscosity for the unsteady turbulent flow past a cylinder: (Left) eddy viscosity ν_t from SST-SAS turbulence model; (Right) the parameter ν_η inferred by the physics-informed neural network.

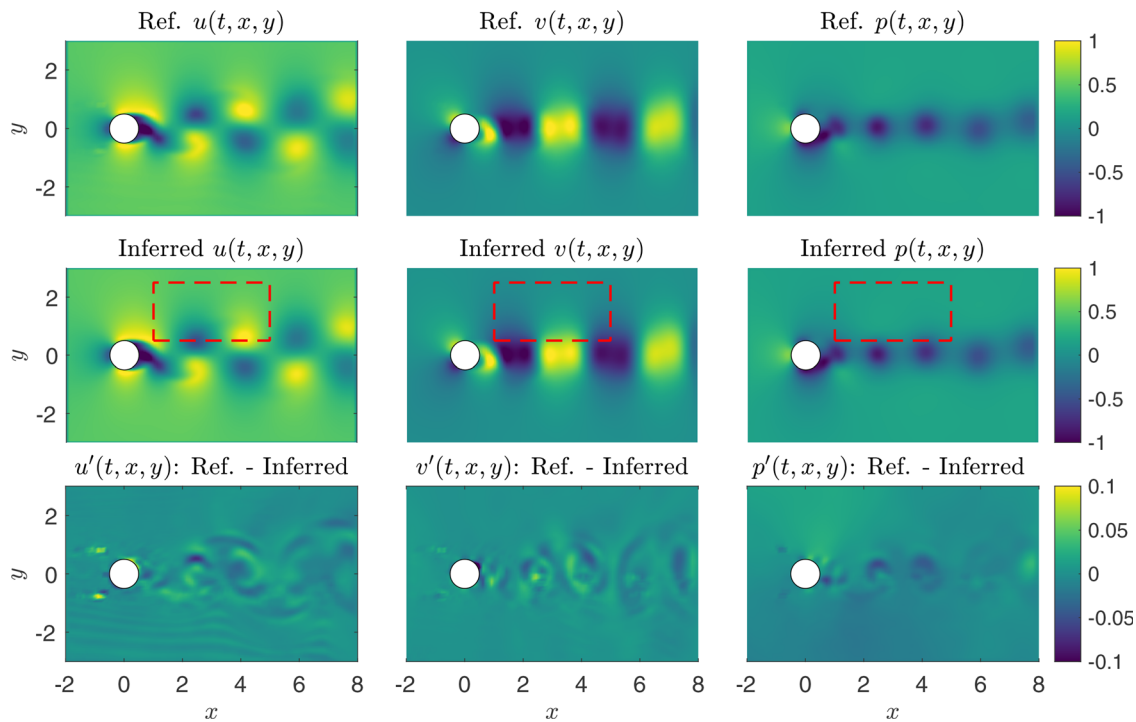


FIG. 11. Unsteady turbulent flow past a cylinder: (Top) the instantaneous reference velocity and pressure fields at $t = 7$, generated by the SST $k - \omega$ turbulence model; (Mid) the inferred fields at the same time instant with the missing dynamics; (Bottom) the difference between the reference fields and the inferred fields. The red dashed-line indicates the region where the flow dynamics is missing.

Significantly, the missing data in a randomly specified region can be inferred by the physics-informed deep learning. Given the reality that reasonable accurate velocity and pressure are predicted by the methodology, the inferred ν_η indicates that the eddy viscosity in the conventional turbulence models probably has an over-dissipation effect on the flow structures and a different spatial distribution from the inferred one. This discovery implies that the physics-preserved data driven techniques may have a transformative influence on the turbulence closure modeling.

C. With white noise

In order to mimic the experimental data and validate the performance of the proposed method for complex spatiotemporal turbulence physics, we consider the situation that the training data are superimposed with 1% and 10% white noise (WN) in a relative sense. The case is kept the same as the one used in Sec. IV B, but without any missing region.

In Fig. 14, the comparison is performed for three cases. Figure 14 (Top) shows the inferred field by the clean numerical simulation by

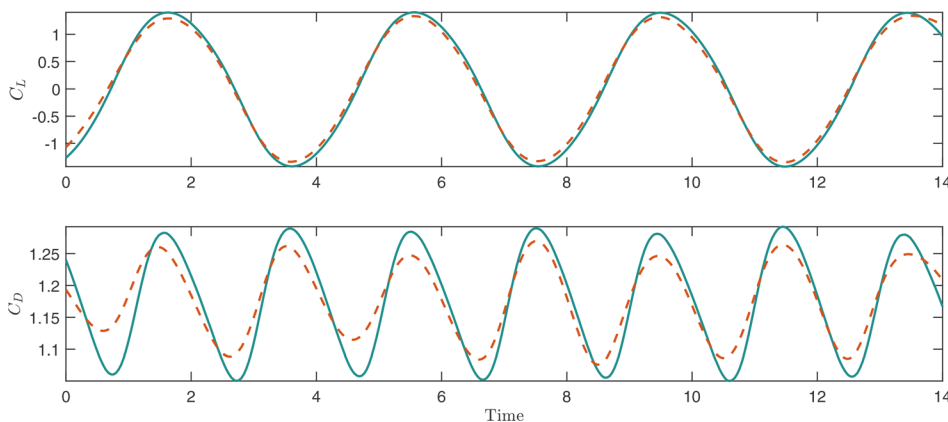


FIG. 12. Lift and drag coefficients over the cylinder: (Top) profiles of the lift coefficient; (Bottom) profiles of the drag coefficient. The solid curves are generated from the reference case by the SST $k - \omega$ turbulence model, while the dashed curves generated from the inferred fields.

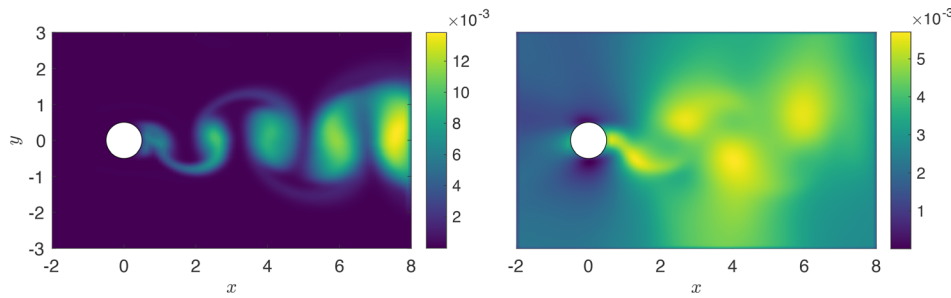


FIG. 13. Comparison between the eddy viscosity and the artificial viscosity for the unsteady turbulent flow past a cylinder: (Left) eddy viscosity ν_t from SST $k-\omega$ turbulence model; (Right) the parameter ν_η inferred by the physics-informed neural network.

the SST-SAS turbulence model, which is treated as a reference field, and Figs. 14(Mid) and 14(Bottom) show the inferred fields by the numerical simulation with added 1% and 10% WN. It is clear that with the current WN magnitudes, the inferred fields present correct unsteady flow patterns. In Fig. 15, the differences between the reference field (0% WN) and the two inferred fields (1% and 10% WN) are presented for a further comparison. From the results, it is obtained that significant differences occur in the wake close to the rear surface of the cylinder. It is demonstrated that with two different noise magnitudes, correct spatiotemporal flow structures can be inferred.

In Fig. 16, the evolutions of the loss function with epochs are depicted with the varying learning rates. It is found that the final values of the loss function are at the same order of magnitude, although the loss function with 1% WN is the smallest. These results validate that

with the background noise, by the training process, consistent convergence can be achieved. Thus, the proposed methodology can be used, with noisy experimental data, to infer correct spatiotemporal flow structures.

In Fig. 17, the profiles of the inferred lift and drag coefficients are given for comparison with the CFD results. Generally, the inferred C_L profiles are consistent with the reference one. Compared with the reference CFD results, the averaged L^2 errors of this dimensionless quantity are 0.66%, 0.52%, and 0.49% for the cases with 0% WN, 1% WN, and 10% WN, respectively. It is clear that the averaged L^2 errors for the lift coefficients are around 0.5%. It is also observed that there exist some larger differences between the inferred C_D and the reference one as presented in Fig. 17. In general, even with the added noise, consistent results with the reference data are obtained.

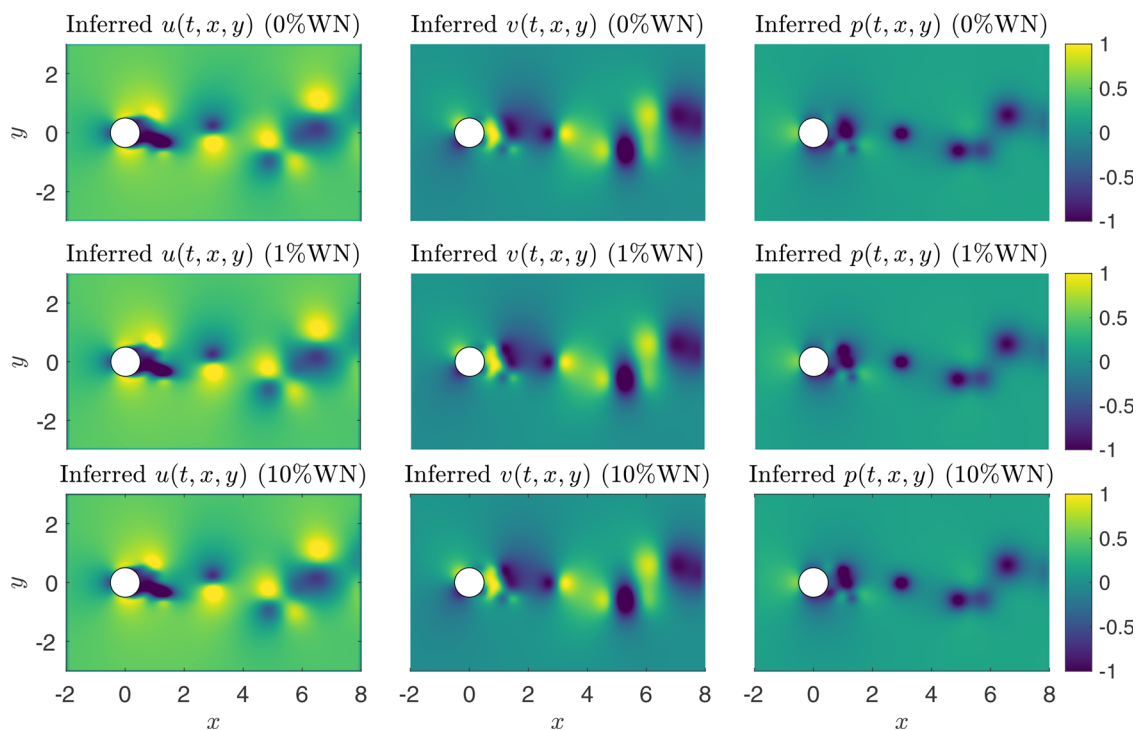


FIG. 14. Comparison of the inferred fields by the clean numerical data and noisy data for the unsteady turbulent flow past a cylinder without missing region: (Top) the inferred whole field by the SST-SAS data; (Mid) the inferred whole field by the SST-SAS data with added 1% WN; (Bottom) the inferred whole field by the SST-SAS data with added 10% WN.

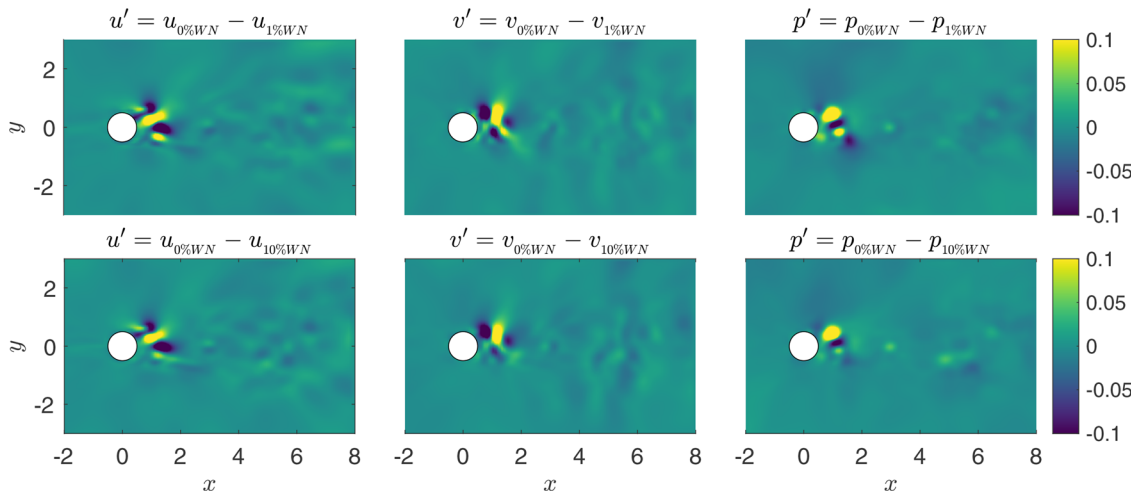


FIG. 15. Differences of the inferred fields by the clean numerical data and noisy data for the unsteady turbulent flow past a cylinder without missing region: (Top) differences between the inferred whole fields without WN and with added 1% WN; (Bottom) differences between the inferred whole fields without WN and with added 10% WN.

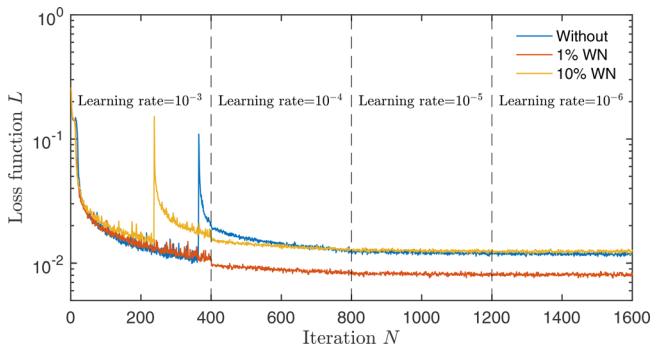


FIG. 16. Loss function evolution for three different cases: (Without) loss function evolution without noise; (1% WN) loss function evolution with added 1% WN; (10% WN) loss function evolution with added 10% WN.

V. CONCLUSION AND DISCUSSION

The methodology proposed in this paper provides a way to explore the missing flow dynamics, which is well validated by the data from the turbulence models. We show that from the single parameterization of the Navier–Stokes equations, the feasibility and potential of the method are significant for the “averaged” or “filtered” data, no matter obtained experimentally (with noise) or numerically. Surprisingly, the current methodology allows to use the parameterized governing equations without extra-modelings, which is shaping a perspective in exploring the dynamics beyond the description of the original governing system, although the deviation of exploring missing dynamics exists. From this point of view, the parameterization, in fact, provides a solid foundation of accurately describing practical complex flow dynamics, which may not be accurately described by the traditional turbulence modelings where the artificial viscosity ν_t needs to be further tuned.

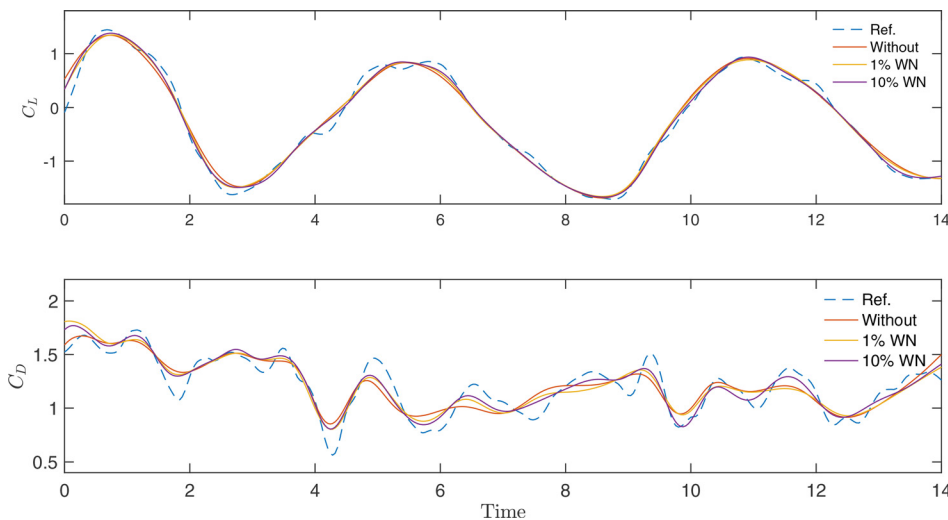


FIG. 17. Comparison of lift and drag coefficients over the cylinder with the CFD reference results: (Top) profiles of lift coefficients; (Bottom) profiles of drag coefficients. Legend labels: (Ref.) from CFD; (Without) loss function evolution without noise; (1% WN) loss function evolution with added 1% WN; (10% WN) loss function evolution with added 10% WN.

The proposed methodology provides a new way for further calibrations of the turbulence closures, which is probably difficult for traditional case-by-case machine learning with simply tuning the scalar modeling parameters. As indicated from the theory of turbulence modeling, due to the significance of the parameter ν_η , the proposed method is naturally termed as “data-driven scale-adaptive turbulent structure recovering.”

It is also interesting to understand how size and location of the missing region influence the results. In this work, we currently only considered two situations, with or without a missing region. It is straightforward that if the missing region equals to the whole domain, without any internal training data, neural network is treated as a tool to solve the governing equations. By studying data assimilation for turbulent flow behind a wall proximity, it is noticed that experimental data in the sensitive region help to improve the inferred results.⁵ This implies that the influence of missing region is flow-dependent, which will be studied in the near future.

Although only 2D incompressible flows were considered as examples in this work, the proposed methodology can be extended to 3D incompressible or compressible flows. While learning from the parameterized Navier–Stokes equations is successful in coping with the traditional fluid dynamics problems, this, in turn, implies that the generalization of the employed methodology could enable exploration and exploitation in a broad range of complex systems, e.g., weather prediction, ocean recirculation, and biofluid mechanics.

AUTHORS' CONTRIBUTIONS

H.X. and W.Z. are co-first authors. They contributed equally to this work.

ACKNOWLEDGMENTS

The authors would like to acknowledge support from the National Numerical Wind Tunnel Project (Grant No. NNW2019ZT4-B09) and the National Natural Science Foundation of China (Grant Nos. 91852106 and 91851127).

The authors have no conflicts to disclose.

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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